APTITUDE

1) A can complete a work in 10 days and B in 15 days. How many days will they take together?

A = 10 days
B = 15 days

$A+B = 1/10 + 1/15$

$= \frac{3+2}{30}$

$\Rightarrow 5/30$

$\boxed{\text{Answer: 6 days}}$

2) A alone can do a work in 12 days after working for 4 days, B joins and work finishes in 4 more days. B's efficiency.

Sol:- A's day work $\Rightarrow 1/12$

A in 4 days:

$4 \times 1/12 = 4/12 = 1/3$

Remaining work $1 - 1/3 = 2/3$

A+B in 4 days.

$\frac{2/3}{4} = 2/12 \Rightarrow 1/6$

$\Rightarrow 1/6 - 1/12$

$\Rightarrow \frac{2-1}{12} \Rightarrow 1/12$

B alone takes 12 days.

$\boxed{\text{12 days}}$

3. If 5 people can complete a work in 20 days how many people are needed to complete the work in 10 days?

Sol..

Men × Days = work

$5 \times 20 = 100$

For 10 days:

Men $\times 10 \Rightarrow 100$

Men $= 10$

$\boxed{\text{10 people}}$ ← Answer

4. A twice as efficient as B. If B takes 12 days to complete work, how many days will A take?

B takes $\Rightarrow 12$ days

$= 1/12$ days

A's efficiency $\Rightarrow \cancel{2} \times (1/\cancel{12}_{6})$

$\Rightarrow 1/6$

A can take 6 days.

$\boxed{\text{6 days}}$

5) Pipe A fills a tank in 6 hours and pipe B fills in 8 hours. How long will both take together?

Sol:-

$$A+B = \frac{1}{6\times4} + \frac{1}{8\times3}$$

$$= \frac{4+3}{24}$$

Reciprocal: $\Rightarrow 7/24$

Time $= 24/7$

$\boxed{\text{Answer: } 24/7 \text{ hours}}$

$\approx$ 3 hours 26 min

Block 2 - VERBAL ABILITY.

1) what is algorithm?

An algorithm is a step-by step procedure used to solve problems efficiently.

2) why are efficient algorithms important?

Efficient algorithms are important because it reduce time complexity and memory usage.

3) what do algorithms help reduce?

They help to reduce time complexity and space complexity.

Grammar Correction

1) ~~The~~ dont like programming

~~the~~ doesn't like programming

2) ~~Ta~~ have written the code

He has written the code.

3. They is working on algorithm.

They are working on algorithm.

Block - 3

COMMUNICATION.

when a program runs, it is first compiled or interpreted, and the operating system allocates memory in RAM for execution.

Then the CPU executes the instructions step by step and finally the program produces output on the screen.